



VP AIR TEST NEET (2025-26)

PART TEST - 10

DROPPER

DURATION: 180 Minutes

Date: 01-03-2026

M. MARKS: 720

Topics Covered

Physics:	Ray Optics and Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor Electronics - Materials, Devices and Simple Circuits
Chemistry:	Coordination Compounds, The p-Block Elements, The d- and f- block Elements, Salt Analysis
Botany:	Plant Growth and Development, Organisms and Population, Ecosystem, Biodiversity and Conservation
Zoology:	Human Health and Disease, Biotechnology - Principles and Processes, Biotechnology and its Applications

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Unpolarized light strikes a transparent medium of refractive index $\sqrt{3}$. If the reflected light is completely plane polarized, the angle of refraction is:

(1) 60° (2) 45°
(3) 30° (4) 90°

2. A nucleus at rest splits into two nuclear parts having velocities in the ratio 8: 1. The ratio of their nuclear radii is:

(1) 1:4 (2) 1:2
(3) 1:8 (4) 4:1

3. Two metals A and B have work functions 3 eV and 5 eV respectively. The ratio of the slopes of the graph plotted between the maximum kinetic energy of emitted photoelectrons (plotted on y-axis) and the frequency of incident radiation (plotted on x-axis) for metal A and B respectively is:

(1) 3: 5 (2) 5:3
(3) 1:1 (4) $\sqrt{3} : \sqrt{5}$

4. Light of wavelength 600 nm is diffracted by a single slit of width 0.2 mm. The pattern is formed on the screen at a distance 100 cm from the slit. The width of the first secondary maximum is:

(1) 2 mm
(2) 3 mm
(3) 6 mm
(4) 1.5 mm

5. In hydrogen-like atoms, how does the ratio of energy difference $(E_{4n} - E_{2n})$ and $(E_{2n} - E_n)$ depend upon atomic number z and principal quantum number n ?

(1) $\frac{z^2}{n^2}$ (2) $\frac{z^4}{n^4}$
(3) $\frac{z}{n}$ (4) $\frac{z^0}{n^0}$

6. Read the following statements:

(A) Volume of the nucleus is directly proportional to the mass number.
(B) Volume of the nucleus is independent of mass number.
(C) Density of the nucleus is directly proportional to the mass number.
(D) Density of nucleus is directly proportional to the cube root of the mass number.

- (E) Density of the nucleus is independent of the mass number.

Choose the **correct** option from the following options.

(1) (A) and (D) only (2) (A) and (E) only.
(3) (B) and (E) only (4) (A) and (C) only.

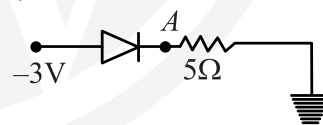
7. The Zener breakdown in a reverse-biased p-n junction is most likely to occur due to:

(1) high velocity of the minority charge carriers if the doping concentration is small
(2) high velocity of the minority charge carriers if the doping concentration is large
(3) strong electric field in the depletion region if the doping concentration is small
(4) strong electric field in the depletion region if the doping concentration is large

8. Two electrons are moving with the same speed v . One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field. After some time, if the de Broglie wavelength of the two are λ_1 and λ_2 , respectively then:

(1) λ_1 must be equal to λ_2
(2) λ_1 must be greater than λ_2
(3) λ_1 may be greater than or less than λ_2
(4) λ_1 must be less than λ_2

9. In the figure given, voltage of point A is: (Assume the diode is ideal)



(1) 0 V (2) -3 V
(3) -2.3 V (4) -2.7 V

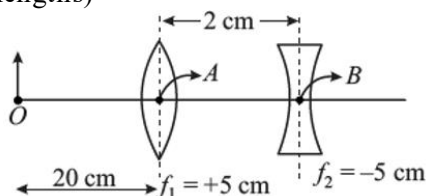
10. In a YDSE apparatus, the slit separation is $d = 1 \text{ mm}$, wavelength of light is $\lambda = 600 \text{ nm}$ and distance of screen from slits is $D = 1 \text{ m}$. Then find the minimum distance between two points on the screen (in mm) where intensity is 75% of the maximum intensity.

(1) 0.1 mm (2) 0.2 mm
(3) 0.3 mm (4) 0.4 mm

11. A convex mirror of radius of curvature 40 cm forms an image that is one-third the size of the object. The object distance is:

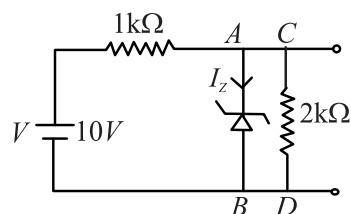
(1) 20 cm (2) 40 cm
(3) 60 cm (4) 30 cm

12. Which of the following statement is **not** true about stopping potential (V_0) in photoelectric experiment?
- It depends on the nature of emitter material.
 - It depends upon frequency of the incident light.
 - It increases with increase in intensity of the incident light.
 - Its magnitude is $1/e$ times the maximum kinetic energy of electrons emitted.
13. When an electron in an hydrogen atom is raised from the ground state to an excited state:
- Both K.E. and P.E. increase
 - Both K.E. and P.E. decrease
 - The P.E. decreases and K.E. increases
 - The P.E. increases and K.E. decreases
14. Given below are two statements:
- Statement I:** In hydrogen atom, the radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2).
- Statement II:** The jumping of electron from higher energy orbit (E_2) to lower energy orbit (E_1) is associated with frequency of radiation given as $f = (E_2 - E_1)/h$
- In the light of the above statements, choose the most appropriate answer from the options given below:
- Both Statement I and Statement II are true
 - Both Statement I and Statement II are false.
 - Statement I is correct but Statement II is false
 - Statement I is incorrect but Statement II is true
15. If an electron of the hydrogen atom is replaced by another particle of same charge but of double the mass, then: (Assume stationary nucleus)
- Bohr radius will increase
 - Ionisation energy of the atom will be doubled
 - Speed of the new particle in a given state will be lesser than the electron's speed in same orbit
 - Gap between energy levels will now be halved
16. What is the position and nature of image formed by lens combination shown in figure? (f_1, f_2 are focal lengths)



- 70 cm left from point B; virtual
- 40 cm right from point B; real
- $20/3$ cm right from point B; real
- 70 cm right from point B; real

17. In the given circuit, the breakdown voltage of the Zener diode is 3.0 V. What is value of I_Z ?



- 3.3 mA
- 5.5 mA
- 10 mA
- 7 mA

18. Given below are two statements:

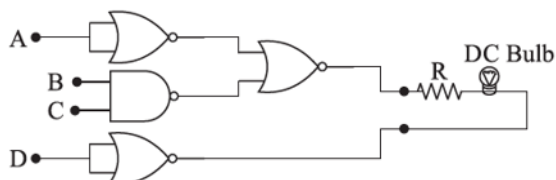
Statement I: When a Silicon sample is doped with Boron, it becomes P type and when doped by Arsenic, it becomes N-type semiconductor.

Statement II: When such P-type and N type semiconductors, are fused to make a junction and reach equilibrium, continuous current will automatically flow which can be detected with an externally connected ammeter.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

19. Following gates section is connected in a complete suitable circuit.



For which of the following combination, bulb will glow (ON)

- $A = 0, B = 1, C = 1, D = 1$
- $A = 1, B = 0, C = 0, D = 0$
- $A = 0, B = 0, C = 0, D = 1$
- $A = 1, B = 1, C = 1, D = 0$

20. The number of beta particles (β^-) emitted by a radioactive substance is twice the number of alpha particles (α) emitted by it. The resulting daughter nucleus is an:

- Isomer of parent
- Isotone of parent
- Isotope of parent
- Isobar of parent

21. The logic circuit shown above is equivalent to:



- (1)
 (2)
 (3)
 (4)

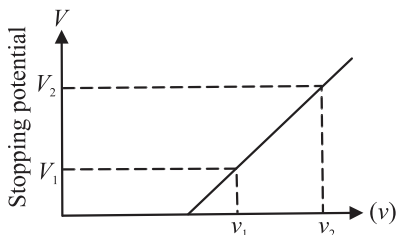
22. An element ${}_Z\text{A}^X$ decays by a two-step process into element C : $A \rightarrow B + {}_2\text{He}^4$, $B \rightarrow C + 2e^-$. Then correct representation of C would be,

- (1) ${}_{Z-2}\text{C}^{X-4}$
 (2) ${}_Z\text{C}^X$
 (3) ${}_{Z-2}\text{C}^X$
 (4) ${}_Z\text{C}^{X-4}$

23. In an n-type semiconductor, which of the following statement is **true**?

- (1) Electrons are minority carriers and pentavalent atoms are dopants.
 (2) Holes are minority carriers and pentavalent atoms are dopants.
 (3) Holes are majority carriers and trivalent atoms are dopants.
 (4) Electrons are majority carriers and trivalent atoms are dopants.

24. Figure shows the graph of stopping potential versus the frequency (ν) of radiation incident on a photosensitive metal. Threshold frequency of the metal is:



- (1) $\frac{V_2\nu_2 + V_1\nu_1}{V_2 + V_1}$ (2) $\frac{V_2\nu_1 + V_1\nu_2}{V_2 + V_1}$
 (3) $\frac{(V_2\nu_1 - V_1\nu_2)}{\nu_2 - \nu_1}$ (4) $\frac{V_2\nu_1 - V_1\nu_2}{V_2 - V_1}$

25. The total energy of an electron in the ground state of a He^+ ion is:

- (1) -13.6 eV (2) -27.2 eV
 (3) -108.8 eV (4) -54.4 eV

26. The photodiode is used to detect optical signals. These diodes are preferably operated in reverse biased mode because:

- (1) Fractional change in majority carriers is not detectable in forward bias.
 (2) Fractional change in minority carriers is not detectable in reverse bias.
 (3) Fractional change in minority carriers is more detectable in forward bias.
 (4) Fractional change in minority carriers is more detectable in reverse bias.

27. The mass numbers of two nuclei are 125 and 64 respectively. The ratio of their nuclear radii is:

- (1) 1.50 (2) 2.25
 (3) 1.25 (4) 1.75

28. A nucleus with mass number 200 has a binding energy per nucleon of 7.4 MeV. It splits into two fragments of equal mass numbers, each having a binding energy per nucleon of 8.2 MeV. The total energy released in this process is:

- (1) 80 MeV (2) 160 MeV
 (3) 240 MeV (4) 120 MeV

29. A plane mirror is moved **3 cm** towards a stationary object. The image will shift by a distance of:

- (1) 9 cm (2) 3 cm
 (3) 6 cm (4) 1.5 cm

30. If the source of light used in a Young's double slit experiment is changed from red to violet:

- (1) The fringes will become brighter.
 (2) Consecutive fringes will come closer
 (3) The central bright fringe will become a dark fringe
 (4) The intensity of minima will increase

31. Two objects A and B are placed in front of concave mirror at 30 cm and 10 cm from the pole of a concave mirror of focal length 20 cm. The distance between the images formed is:

- (1) 40 cm (2) 80 cm
 (3) 20 cm (4) 60 cm

32. Light enters a medium from air at an angle of incidence of 60° . If angle of deviation is 30° after refraction, the refractive index of the medium is:

- (1) 1.5 (2) $\sqrt{2}$
 (3) 1.33 (4) $\sqrt{3}$

33. An equilateral prism is made of a material of refractive index $\sqrt{2}$. The angle of minimum deviation is:
 (1) 45° (2) 60°
 (3) 30° (4) 90°
34. A light ray falls on a medium-air interface at an angle of incidence of 45° . For which of the following value of refractive index μ will the ray undergo Total Internal Reflection (TIR)?
 (1) 1.3 (2) 1.5
 (3) 1.1 (4) 1.2
35. Light from a source at the bottom of a liquid of depth $\sqrt{3}$ cm emerges and just grazes the surface at a horizontal radius of 1 cm. The speed of light in the liquid is:
 (1) 2.0×10^8 m/s (2) 1.5×10^8 m/s
 (3) 2.5×10^8 m/s (4) 1.2×10^8 m/s
36. In a Young's double slit experiment, the intensity at the central maximum is I_0 . The intensity at a point where the path difference is $\frac{\lambda}{4}$ will be:
 (λ is the wavelength of light used):
 (1) I_0 (2) $\frac{I_0}{2}$
 (3) $\frac{I_0}{4}$ (4) $\frac{I_0}{\sqrt{2}}$
37. Two coherent sources have an amplitude ratio of 2:1. The ratio of maximum intensity to minimum intensity ($I_{\max} : I_{\min}$) in the interference pattern is:
 (1) 4:1 (2) 3:1
 (3) 9:1 (4) 2:1
38. In YDSE, the slit separation is 0.5 mm and the screen is 0.5 m away. If the first dark fringe is formed on the screen directly in front of one of the slits, the wavelength of light used is:
 (1) 450 nm (2) 600 nm
 (3) 550 nm (4) 500 nm
39. An electron in a hydrogen atom is in an excited state with a total energy of -1.51 eV. The de-Broglie wavelength of this electron is approximately:
 (1) 3.3 Å (2) 10 Å
 (3) 3.6 Å (4) 12.27 Å
40. Unpolarized light of intensity 32 W/m^2 passes through three polaroids. The first and last polaroids are crossed (axes perpendicular). If the emergent intensity is 1 W/m^2 , the angle between the pass axes of the first two polaroids is:
 (1) 45° (2) 30°
 (3) 15° (4) 60°
41. For a photosensitive surface, incident light of wavelength λ gives a stopping potential of $3V$, and wavelength 2λ gives $1V$. The threshold wavelength for this surface is:
 (1) 3λ (2) 4λ
 (3) 2.5λ (4) 5λ
42. The ratio of the wavelength of photon emitted when an electron in a Hydrogen atom jumps from $n=3 \rightarrow n=1$ to the wavelength emitted for $n=2 \rightarrow n=1$ is:
 (1) 32:27 (2) 27:32
 (3) 9:4 (4) 8:3
43. A pure semiconductor has:
 (1) An infinite resistance at 0°C
 (2) A finite resistance which does not depend upon temperature
 (3) A finite resistance which increases with increase in temperature
 (4) A finite resistance which decreases with increase in temperature
44. A nuclear reactor delivers a power of 900 MW. If the energy is generated by mass-energy conversion, the mass of fuel converted into energy per hour is:
 (1) 10 mg (2) 40 mg
 (3) 36 mg (4) 20 mg
45. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\frac{\lambda}{2}$ separately. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function (ϕ) of the surface of the material is: (h = Planck's constant, c = speed of light)
 (1) $\frac{hc}{3\lambda}$ (2) $\frac{hc}{2\lambda}$
 (3) $\frac{hc}{\lambda}$ (4) $\frac{2hc}{\lambda}$

46. Excess AgNO_3 is added separately to the complexes:

- A. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 B. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 C. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

Only the ionisable (outer-sphere) chloride ions react with Ag^+ to form AgCl(s) . Arrange the complexes A, B and C in the decreasing order of moles of AgCl formed.

- (1) $A > B > C$ (2) $C > B > A$
 (3) $B > C > A$ (4) $C > A > B$

47. The highest enthalpy of atomisation among the following is for:

- (1) Ti (2) V
 (3) Cr (4) Zn

48. Which group 13 element has the highest density (298 K)?

- (1) Al (2) Ga
 (3) In (4) Tl

49. The square planar complex $[\text{Pt}(\text{NH}_3)(\text{py})\text{ClI}]$ has how many geometrical isomers?

- (1) 1 (2) 2
 (3) 3 (4) 4

50. Which element has the most negative $E^\circ (\text{M}^{2+}/\text{M})$?

- (1) Ti (2) Fe
 (3) Cu (4) Zn

51. Which has the lowest melting point among group 13 elements?

- (1) B (2) Al
 (3) Ga (4) In

52. **Correct** increasing order of covalent radius is:

- (1) $\text{C} < \text{Si} < \text{Ge} < \text{Sn} < \text{Pb}$
 (2) $\text{Si} < \text{C} < \text{Ge} < \text{Sn} < \text{Pb}$
 (3) $\text{C} < \text{Ge} < \text{Si} < \text{Sn} < \text{Pb}$
 (4) $\text{C} < \text{Si} < \text{Sn} < \text{Ge} < \text{Pb}$

53. Which complex is expected to show optical isomerism?

- (1) $\text{trans-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 (2) $\text{cis-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (4) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

54. Which ion has the smallest ionic radius among:

- (1) Fe^{2+} (2) Co^{2+}
 (3) Ni^{2+} (4) Cu^{2+}

55. Which catalyst is used in the Haber process for the production of ammonia from N_2/H_2 mixture?

- (1) PdCl_2
 (2) TiCl_4 with $\text{Al}(\text{C}_2\text{H}_5)_3$
 (3) Iron catalysts
 (4) V_2O_5

56. Which element has the highest first ionisation enthalpy ($\Delta_f H_1$)?

- (1) Si
 (2) Ge
 (3) Pb
 (4) C

57. Match the **List-I** with **List-II** (spin-only magnetic moments):

List-I		List-II	
(A)	$[\text{Fe}(\text{CN})_6]^{4-}$	(I)	5.92 BM
(B)	$[\text{FeF}_6]^{3-}$	(II)	0 BM
(C)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(III)	4.90 BM
(D)	$[\text{Fe}(\text{CN})_6]^{3-}$	(IV)	1.73 BM

Choose the **correct** answer from the options given below;

- (1) A-II, B-I, C-III, D-IV
 (2) A-I, B-II, C-III, D-IV
 (3) A-II, B-III, C-I, D-IV
 (4) A-III, B-I, C-II, D-IV

58. Which oxide corresponds to the highest oxidation state shown for 3d metals in the table?

- (1) CrO_3
 (2) MnO_2
 (3) Mn_2O_7
 (4) Fe_2O_3

59. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: First ionisation enthalpy decreases from N to Bi in group 15.

Reason R: Atomic size increases down the group, so the outer electron is held less strongly.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
 (2) Both A and R are true but R is NOT the correct explanation of A.
 (3) A is true but R is false.
 (4) A is false but R is true.

60. **Correct** increasing order of ligand field strength is:
- (1) $\text{CN}^- < \text{NH}_3 < \text{H}_2\text{O} < \text{F}^-$
 - (2) $\text{F}^- < \text{H}_2\text{O} < \text{NH}_3 < \text{CN}^-$
 - (3) $\text{H}_2\text{O} < \text{F}^- < \text{NH}_3 < \text{CN}^-$
 - (4) $\text{NH}_3 < \text{H}_2\text{O} < \text{F}^- < \text{CN}^-$
61. Oxygen stabilises very high oxidation states better than fluorine mainly because oxygen:
- (1) has larger size, so it makes weaker bonds
 - (2) forms multiple bonds with metals ($p\pi-d\pi$), giving extra stability
 - (3) is less electronegative than fluorine so it always reduces metals
 - (4) cannot form bridges between metal atoms
62. Which oxide is prepared by: $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$ (cooling/condensation)?
- (1) N_2O
 - (2) NO
 - (3) N_2O_4
 - (4) N_2O_5
63. For an octahedral complex of a metal ion, $\Delta_0 = 18000 \text{ cm}^{-1}$. Approximate Δ_t for the tetrahedral complex of the same metal-ligand pair is:
- (1) 4000 cm^{-1}
 - (2) 6000 cm^{-1}
 - (3) 8000 cm^{-1}
 - (4) 12000 cm^{-1}
64. Which pair has same oxidation state of phosphorus?
- (1) H_3PO_3 and $\text{H}_4\text{P}_2\text{O}_5$
 - (2) H_3PO_2 and H_3PO_4
 - (3) $\text{H}_4\text{P}_2\text{O}_6$ and $\text{H}_4\text{P}_2\text{O}_7$
 - (4) H_3PO_4 and H_3PO_3
65. Which halogen molecule has the maximum bond dissociation enthalpy (kJ mol^{-1}) among those listed?
- (1) F_2
 - (2) Cl_2
 - (3) Br_2
 - (4) I_2
66. Complete hydrolysis of XeF_2 gives:
- (1) $\text{Xe} + \text{HF}$ only
 - (2) $\text{Xe} + \text{O}_2$ only
 - (3) $\text{Xe} + \text{HF} + \text{O}_2$
 - (4) $\text{XeO}_3 + \text{HF}$
67. Total stereoisomers possible for $[\text{Co}(\text{en})_2\text{Br}_2]^+$ are:
- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4
68. $\text{Cu}(\text{II})$ iodide is **not** known because:
- (1) I^- oxidises Cu^{2+} to Cu^{3+}
 - (2) Cu^{2+} oxidises I^- to I_2 and forms CuI
 - (3) Cu^{2+} forms only covalent bonds with iodide
 - (4) CuI oxidises I^- to I_2
69. In aqueous (acidic) solution, Al^{3+} forms a complex A. Identify A and hybridisation of Al in A.
- (1) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$, sp^3d^2
 - (2) $[\text{AlCl}_6]^{3-}$, d^2sp^3
 - (3) $[\text{Al}(\text{H}_2\text{O})_4]^{3+}$, sp^3
 - (4) $[\text{Al}(\text{OH})_6]^{3-}$, sp^3d^2
70. Hydrolysis of XeF_6 leads to formation of XeOF_4 and XeO_2F_2 . This hydrolysis is:
- (1) a redox reaction because Xe changes oxidation state
 - (2) a redox reaction because F changes oxidation state
 - (3) not a redox reaction because oxidation states remain unchanged
 - (4) not a redox reaction because only HF is formed
71. The close similarity in radii of Zr and Hf (hence difficulty in separation) is mainly due to:
- (1) shielding effect of 3d electrons
 - (2) lanthanoid contraction
 - (3) actinoid contraction only
 - (4) inert pair effect
72. Which statement is **correct** regarding formation of fluoroanions?
- (1) XeF_2 reacts with PF_5 to give $[\text{XeF}]^+[\text{PF}_6]^-$
 - (2) XeF_4 reacts with SbF_5 to give $[\text{XeF}_3]^+[\text{SbF}_6]^-$
 - (3) XeF_6 reacts with alkali metal fluoride MF to give $\text{M}^+[\text{XeF}_7]^-$
 - (4) XeF_6 reacts with water to give $[\text{XeF}_7]^-$
73. Arrange in increasing wavelength of absorption (visible region): $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{CoF}_6]^{3-}$
- (1) $[\text{CoF}_6]^{3-} < [\text{Co}(\text{H}_2\text{O})_6]^{3+} < [\text{Co}(\text{NH}_3)_6]^{3+}$
 - (2) $[\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{H}_2\text{O})_6]^{3+} < [\text{CoF}_6]^{3-}$
 - (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+} < [\text{CoF}_6]^{3-} < [\text{Co}(\text{NH}_3)_6]^{3+}$
 - (4) $[\text{Co}(\text{NH}_3)_6]^{3+} < [\text{CoF}_6]^{3-} < [\text{Co}(\text{H}_2\text{O})_6]^{3+}$
74. Which of the following aqueous ions is expected to be colourless?
- (1) Ti^{3+}
 - (2) V^{3+}
 - (3) Zn^{2+}
 - (4) Cu^{2+}
75. Brown ring in nitrate test is due to formation of:
- (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 - (2) $[\text{Fe}(\text{CN})_6]^{4-}$
 - (3) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$
 - (4) $\text{Fe}(\text{NO}_3)_3$

76. Which of the following ions is paramagnetic?
 (1) La^{3+} ($4f^0$) (2) Lu^{3+} ($4f^{14}$)
 (3) Ce^{3+} ($4f^1$) (4) Both (1) and (2)
77. Which octahedral complex shows zero CFSE?
 (1) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{CoF}_6]^{3-}$ (4) $[\text{Fe}(\text{CN})_6]^{4-}$
78. Salt treated with dilute acid gives a gas with rotten egg smell and it blackens lead acetate paper. Anion is:
 (1) S^{2-} (2) SO_4^{2-}
 (3) Cl^- (4) NO_3^-
79. Which ion is expected to be colourless because it has no f-f transitions?
 (1) Ce^{3+} (2) Eu^{2+}
 (3) La^{3+} (4) Nd^{3+}
80. Which of the following is an inner-orbital complex and also diamagnetic?
 (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{CoF}_6]^{3-}$
 (3) $[\text{Ni}(\text{CN})_4]^{2-}$ (4) $[\text{Zn}(\text{NH}_3)_4]^{2+}$
81. Given below are two statements:
Statement I: In acidic medium, chromate ions convert to dichromate ions.
Statement II: In basic medium, dichromate ions convert to chromate ions.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.
82. Passing H_2S in acidic medium precipitates sulphides of group II cations. Which cation belongs to this group?
 (1) Fe^{3+} (2) Zn^{2+}
 (3) Pb^{2+} (4) Na^+
83. Heating a salt with NaOH gives a gas that turns red litmus blue and forms white fumes with HCl . Cation is:
 (1) Na^+
 (2) NH_4^+
 (3) Ca^{2+}
 (4) Al^{3+}
84. Among the following, identify the hybridisation of Ni in $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ respectively:
 (1) sp^3 , sp^3 (2) dsp^2 , sp^3
 (3) sp^3 , dsp^2 (4) d^2sp^3 , sp^3
85. The most stable/common oxidation state shown by lanthanoids is:
 (1) +1 (2) +2
 (3) +3 (4) +6
86. Which of the following statements are **correct** for sulphate ion test?
 A. SO_4^{2-} forms a white precipitate with BaCl_2 solution in acidic medium.
 B. The precipitate of BaSO_4 is soluble in dilute HCl .
 C. BaSO_4 is highly insoluble in water and dilute acids.
 D. Acidification with HCl prevents interference from carbonate ions.
 (1) A, C and D only (2) A and B only
 (3) B and C only (4) A and D only
87. A salt gives brown fumes on adding dilute acid, and also turns starch-iodide paper blue. Anion is:
 (1) NO_3^- (2) NO_2^-
 (3) Cl^- (4) CH_3COO^-
88. The coordination number of Ni in $[\text{Ni}(\text{en})_2\text{Cl}_2]$ is: (en = ethylenediamine, bidentate)
 (1) 4 (2) 5
 (3) 6 (4) 2
89. Given below are two statements:
Statement I: Manganate ion and permanganate ion are tetrahedral.
Statement II: Green manganate ion is diamagnetic due to absence of unpaired electrons.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.
90. A solution of KMnO_4 oxidises iodide such that the product is iodate (IO_3^-) and manganese ends up as MnO_2 . The medium is most likely:
 (1) strongly acidic
 (2) neutral or faintly alkaline
 (3) strongly alkaline only
 (4) any medium (product same)

91. Which of the following is/are **not** an attribute of a population?

- (1) Birth rate (2) Death rate
(3) Sex ratio (4) Births and deaths

92. Match List-I with List-II.

List-I		List-II	
(A)	Auxin	(I)	Hastens the maturity period in juvenile conifers
(B)	Gibberellin	(II)	Promotes female flowers in cucumbers
(C)	Cytokinin	(III)	Induces parthenocarpy in tomatoes
(D)	Ethylene	(IV)	Promotes nutrient mobilisation which helps in delay of leaf senescence

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
(2) A-III, B-II, C-I, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-II, B-IV, C-I, D-III

93. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The Abingdon tortoise in Galapagos Islands became extinct within a decade after goats were introduced on the island.

Reason (R): Gause's 'Competitive Exclusion Principle' states that two closely related species competing for different resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

94. Arrange the following groups found in Amazonian rain forest in the descending order of their species count.

- A. Birds
B. Mammals
C. Plants
D. Fishes
E. Reptiles

- (1) D > A > B > C > E
(2) C > D > A > B > E
(3) D > A > C > B > E
(4) A > D > C > B > E

95. Identify the **true** statement about decomposition.

- (1) It is slower if detritus is rich in nitrogen and sugars.
(2) It is quicker if detritus is rich in lignin and chitin.
(3) Warm and moist environment inhibit decomposition.
(4) Low temperature and anaerobiosis inhibit decomposition.

96. Given below are two statements:

Statement I: The exponential growth of an organism can be expressed as $W_1 = W_0 e^{rt}$, where 'r' is the absolute growth rate.

Statement II: In arithmetic growth, following mitotic cell division, both daughter cells continue to divide, while in geometric growth only one matures and differentiates.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

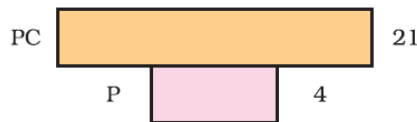
97. Read the following statements.

- A. Ecology gives us a holistic perspective to biology.
B. The essence of biological understanding is to know how organisms interact with other organisms and physical habitats as a group.
C. A particular aspect of ecology is the study of anthropogenic environmental degradation.
D. Ramdeo Misra is revered as the Father of Ecology in India.
E. Ramdeo Misra was awarded the Nobel prize in 1972.

How many of the above statements are **correct**?

- (1) Two only
(2) Three only
(3) Four only
(4) Five

98. The given figure represents an ecological pyramid in a particular ecosystem. Choose the **true** statement about it.



- (1) The biomass of fishes far exceeds that of phytoplankton.
 (2) It represents the pyramid of energy.
 (3) This is a typical pyramid of number in an aquatic ecosystem.
 (4) Such a pyramid of biomass is typically found in grassland ecosystem.
99. Which of the following is **not** one of the three biodiversity hotspots of India?
 (1) Western Ghats & Sri Lanka
 (2) Indo-Burma
 (3) Chanda and Bastar areas
 (4) Himalaya
100. In a grassland ecosystem, if the energy available at the producer level is 10,000 J, what would be the energy available to the secondary consumer?
 (1) 1000 J (2) 100 J
 (3) 10 J (4) 1 J
101. Who reported the appearance of symptoms of 'bakanae' disease in rice seedlings when they were treated with sterile filtrates of the fungus *Gibberella fujikuroi*?
 (1) F. Skoog
 (2) Charles Darwin
 (3) E. Kurosawa
 (4) F.W. Went
102. Identify the **incorrectly** matched pair.
 (1) Mutualism : Both species are benefitted
 (2) Commensalism : One species benefits, the other is neither harmed nor benefitted
 (3) Amensalism : One species is harmed whereas the other is unaffected
 (4) Competition : One species loses while the other is unaffected
103. The passenger pigeon became extinct due to;
 (1) over-exploitation.
 (2) habitat loss and fragmentation.
 (3) co-extinction.
 (4) alien-species invasion.

104. The percentage of incident solar radiation that is considered as photosynthetically active radiation is;
 (1) more than 70 per cent.
 (2) less than 50 per cent.
 (3) 100 per cent.
 (4) 90 per cent.
105. Removal of the apical bud results in the growth of lateral buds. This is widely applied in:
 (1) increasing the length of grapes stalks.
 (2) flowering in pineapple.
 (3) tea plantations and hedge-making.
 (4) preparing weed-free lawns.
106. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The ecological pyramids do not include saprophytes.
Reason (R): Saprophytes play a vital role in the ecosystem.
 In the light of the above statements, choose the *most appropriate* answer from the options given below.
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) (A) is false but (R) is true.
107. A population had 30 pre-reproductive, 40 reproductive and 20 post-reproductive individuals. Such a population is said to be;
 (1) stable.
 (2) declining.
 (3) increasing.
 (4) stationary.
108. On a logarithmic scale, the relationship between species-richness and area is;
 (1) rectangular hyperbola.
 (2) J-shaped curve.
 (3) sigmoid curve.
 (4) straight line.
109. The **correct** set of synthetic auxins is;
 (1) IAA and IBA.
 (2) 2, 4-D and NAA.
 (3) IAA and 2, 4-D.
 (4) 2, 4-D and IBA.

110. Match List-I with List-II.

List-I (Organism)		List-II (Recently extinct from)	
(A)	Dodo	(I)	Australia
(B)	Quagga	(II)	Russia
(C)	Thylacine	(III)	Mauritius
(D)	Steller's Sea Cow	(IV)	Africa

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-II, C-I, D-IV
- (4) A-I, B-IV, C-II, D-III

111. The yield of sugarcane can be increased by spraying their crop with;

- (1) auxins.
- (2) gibberellins.
- (3) cytokinins.
- (4) ethephon.

112. A man-made aquatic ecosystem is:

- (1) estuary.
- (2) wetland.
- (3) pond.
- (4) aquarium.

113. Given below are two statements:

Statement I: A predator that is too efficient and overexploits its prey will eventually become extinct.

Statement II: To lessen the impact of predation, prey species have evolved defences like camouflage and production of chemicals.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

114. The ability of plants to follow different pathways in response to the environment or phases of life to form different kinds of structures is called:

- (1) development.
- (2) plasticity.
- (3) differentiation.
- (4) redifferentiation.

115. In a pond ecosystem, the decomposers are;

- A. Flagellates.
- B. Algae.
- C. Fungi.
- D. Zooplanktons.
- E. Bacteria.

Choose the **correct** option.

- (1) A, B and D
- (2) B, D and E
- (3) A, C and E
- (4) B, C and D

116. The magnitude of “intrinsic rate of natural increase” for the flour beetle is;

- (1) 1.5
- (2) 0.0205
- (3) 0.015
- (4) 0.12

117. Which of the following physiological effects are regulated by Ethylene?

- A. Promotes senescence and abscission of leaves and flowers.
- B. Breaks seed and bud dormancy.
- C. Hastens fruit ripening in tomatoes and apples.
- D. Promotes root growth and root hair formation.
- E. Stimulates stomatal closure during water stress.

Choose the **correct** option.

- (1) A, B, C and D
- (2) B, C, D and E
- (3) A, C, D and E
- (4) A, B, D and E

118. Identify the **correct statement.**

- (1) According to IUCN (2004), the total number of plant and animal species described is slightly less than 1.5 million.
- (2) Robert May places the global species diversity at about 7 million.
- (3) More than 70 per cent of all species recorded are plants.
- (4) The number of fungi species is less than that of amphibians.

119. Select the **incorrectly matched pair.**

- (1) Crow - Omnivore
- (2) Phytoplankton - Producer
- (3) Algae - Herbivore
- (4) Cockroach - Omnivore

120. Pacific salmon fish and bamboo:

- (1) breed many times during their lifetime.
- (2) have a very low reproductive fitness.
- (3) breed only once in their lifetime.
- (4) have evolved towards least effective reproductive strategy.

121. A root elongating at a constant rate;

- (1) shows geometric growth.
- (2) has a sigmoid curve when growth is plotted against time.
- (3) has a J-shaped curve when growth is plotted against time.
- (4) shows arithmetic growth.

122. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In general, species diversity decreases as we move away from the equator towards the poles.

Reason (R): Tropical environments, unlike temperate ones, are more seasonal, less constant and unpredictable.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

123. The humus;

- (1) is the raw material for decomposition.
- (2) is a colloidal substance.
- (3) lacks nutrients.
- (4) cannot be degraded further.

124. Match List-I with List-II.

List-I		List-II	
(A)	Predation	(I)	Bird eating the seeds
(B)	Parasitism	(II)	Ticks on dogs
(C)	Competition	(III)	Visiting flamingoes and resident fishes in some South American lakes
(D)	Commensalism	(IV)	Barnacles on whale

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-II, D-I
- (4) A-I, B-II, C-III, D-IV

125. Read the following statements about productivity.

- A. The rate of biomass production is called productivity.
- B. Gross primary productivity minus respiration losses is the net primary productivity.
- C. Secondary productivity is the rate of formation of new organic matter by producers.
- D. The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight).
- E. The productivity of oceans is 55 billion tons despite occupying 70 per cent of the surface.

The **incorrect** statement(s) is/are;

- (1) A and C.
- (2) E only.
- (3) C only.
- (4) B and D

126. If 4 individuals in a laboratory population of 40 fruitflies died during a week, what is the death rate in the population during that period?

- (1) 0.1 individuals per fruitfly per week
- (2) 10 individuals per fruitfly per week
- (3) 0.4 individuals per fruitfly per week
- (4) 1.0 individuals per fruitfly per week

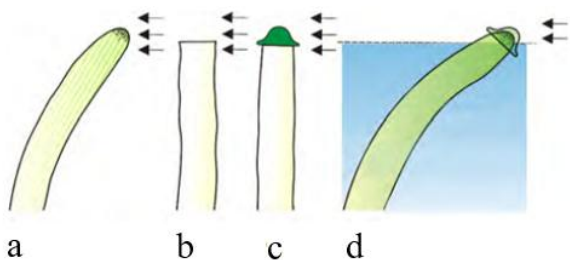
127. In most situations, ABA acts as an antagonist to;

- (1) auxins.
- (2) ethylene.
- (3) cytokinins.
- (4) gibberellic acids.

128. David Tilman's long-term ecosystem experiments using outdoor plots showed that;

- (1) increased diversity contributed to lower productivity.
- (2) plots with more species showed less year-to-year variation in total biomass.
- (3) biodiversity had no impact on productivity.
- (4) increased diversity leads to resource partitioning.

129. In the experiment shown below, the material used for investigation was;



- (1) callus of tobacco.
 (2) tip of coleoptile of oat seedling.
 (3) tip of the coleoptile of canary grass.
 (4) tip of deep water rice plants.
130. Identify the correct statements from the following.
- A. Natural selection operates at individual level.
 B. All the cormorants in a wetland constitute a population.
 C. Population ecology links ecology to population genetics and evolution.
 D. A group of individuals resulting from asexual reproduction is not considered a population.
 E. In nature, we rarely find isolated, single individuals of any species.

Choose the **correct** option.

- (1) B, C, E
 (2) A, B, C
 (3) C, D, E
 (4) A, D, E
131. Given below are two statements:
- Statement I:** Saving the entire forest to save the tiger is an example of *in situ* conservation.
Statement II: When an animal is endangered and needs urgent measures to save it from extinction, *in situ* conservation is the desirable approach.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

132. Which of the following is **not** a functional feature of an ecosystem?

- (1) Species composition
 (2) Decomposition
 (3) Energy flow
 (4) Nutrient cycling

133. The phenomenon by which cork cambium is formed from fully differentiated parenchyma cells is called;

- (1) differentiation.
 (2) redifferentiation.
 (3) heterophylly.
 (4) dedifferentiation.

134. If a new habitat is just being colonised, which of the following will most significantly contribute to increase in population?

- (1) Natality
 (2) Immigration
 (3) Mortality
 (4) Emigration

135. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Fragmentation	(I)	Formation of dark colored amorphous substance
(B)	Leaching	(II)	Water-soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts
(C)	Catabolism	(III)	Break down of detritus into smaller particles
(D)	Humification	(IV)	Bacterial and fungal enzymes degrade detritus into simpler inorganic substances

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
 (2) A-IV, B-II, C-I, D-III
 (3) A-III, B-II, C-IV, D-I
 (4) A-I, B-III, C-II, D-IV

136. Which of the following is **not** a function of the innate immune system barriers?

- (1) Skin prevents entry of microorganisms.
- (2) Lysozyme in tears and saliva destroy bacterial cell walls.
- (3) Production of specific antibodies against particular pathogens.
- (4) Stomach acid denatures ingested pathogens.

137. Match List-I with List-II.

List-I		List-II	
(A)	Cotton bollworm	(I)	Produce insecticidal protein crystals during a particular phase of their growth
(B)	Corn borer	(II)	<i>CryIAb</i> gene
(C)	Tobacco infected with nematodes	(III)	dsRNA interference
(D)	<i>Bacillus thuringiensis</i>	(IV)	<i>CryIAC</i> gene

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-IV, B-II, C-I, D-III
- (3) A-II, B-IV, C-III, D-I
- (4) A-I, B-III, C-II, D-IV

138. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Restriction enzymes always cut DNA to produce blunt ends.

Reason R: Restriction enzymes recognize palindromic sequences and cut both strands at specific distances from the center.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is true, but R is false.
- (4) A is false, but R is true.

139. Consider the following statements about HIV/AIDS:

- A. HIV belongs to the family of retroviruses with an RNA genome
- B. The virus preferentially attacks CD₄⁺ helper T-lymphocytes
- C. HIV can be transmitted through saliva and sweat of infected persons
- D. The diagnostic test for AIDS uses ELISA based on antigen-antibody interaction
- E. Anti-retroviral drugs can completely cure AIDS when given early

Which of the above statements are **correct**?

- (1) B, C, D and E only
- (2) A, B, C and D only
- (3) A, B, C, D and E
- (4) A, B and D only

140. A pharmaceutical company wanted to produce human insulin on a large scale using recombinant DNA technology. After successfully cloning the insulin A and B chains separately in *E. coli*, the major technical challenge they faced was:

- (1) Expressing the genes in mammalian cell lines.
- (2) Creating the correct disulfide bonds between A and B chains.
- (3) Removing the signal peptide from the protein.
- (4) Scaling up the production in fermentation tanks.

141. Given below are two statements:

Statement I: Macrophages and neutrophils are part of the adaptive immune system and produce specific antibodies.

Statement II: Innate immune system provides the first line of defense with non-specific cellular responses.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

142. In the PCR (Polymerase Chain Reaction), which of the following is **not** essential for the process?

- (1) *Taq* polymerase from *Thermus aquaticus*
- (2) Primers complementary to target DNA sequences
- (3) deoxyribonucleotides (dNTPs)
- (4) Restriction enzymes to cut the amplified DNA

143. Match the List-I with List-II:

List-I		List-II	
(A)	pBR322	(I)	Molecular scissors
(B)	Restriction enzymes	(II)	Plasmid with amp ^R and tet ^R genes
(C)	Ti plasmid	(III)	From <i>Agrobacterium tumefaciens</i>
(D)	DNA ligase	(IV)	Molecular glue

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-II, B-I, C-III, D-IV

144. Which of the following is **incorrectly** paired?
- (1) Malaria - *Plasmodium falciparum* - Female *Anopheles* mosquito
 - (2) Typhoid - *Salmonella typhi* - Contaminated food and water
 - (3) Filariasis - *Wuchereria bancrofti* – Sand fly
 - (4) Amoebiasis - *Entamoeba histolytica* - Large intestine infection

145. Consider the following statements about tissue culture and plant biotechnology:

- A. Totipotency allows a single plant cell to regenerate an entire plant.
- B. Micropropagation produces genetically identical somaclones.
- C. Explants must be cultured in medium with only mineral salts.
- D. Meristem culture can produce virus-free plants from diseased material.
- E. Somatic hybridization requires protoplast fusion between two plant cells.

Which of the above statements is/are **not** correct?

- (1) A, B, D and E only
- (2) A, B, C, D and E
- (3) C only
- (4) C and E only

146. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Thymus gland is responsible for producing T-lymphocytes and is quite large at birth but reduces to a very small size by puberty.

Reason R: The thymus produces thymosins which promote B-cell maturation and development of cell-mediated immunity.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both A and R are true, and R is the correct explanation of A
- (2) Both A and R are true, but R is not the correct explanation of A
- (3) A is true, but R is false
- (4) A is false, but R is true

147. Which of the following statements about *Bt* toxin is **correct**?

- (1) *Bt* toxin is active in all insects including aphids and scale insects.
- (2) *Bt* toxin proteins are active in the *Bacillus thuringiensis* gut because of the acidic pH.
- (3) *Bt* toxin works by binding to midgut epithelial cells and forming pores in the intestinal wall.
- (4) All transgenic *Bt* plants produce identical toxins regardless of species.

148. Match List-I with List-II.

List-I (Pathogenic organism)		List-II (Mode of transmission)	
(A)	<i>Trichophyton</i>	(I)	Droplets from infected person
(B)	Rhino virus	(II)	Soil and contaminated clothes
(C)	<i>Wuchereria bancrofti</i>	(III)	Housefly as mechanical carriers
(D)	<i>Entamoeba histolytica</i>	(IV)	Female <i>Culex</i> mosquito

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-II, B-I, C-III, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-IV, B-I, C-II, D-III

149. Consider the following four statements (A-D) and select the **correct** option stating which ones are true (T) and which ones are false (F).

- A. Antibodies are produced by T-lymphocytes.
- B. Cell-mediated immunity is provided by B-lymphocytes.
- C. IgG antibodies provide long-term immunity.
- D. Bone marrow is where all blood cells including lymphocytes are produced.

	A	B	C	D
(1)	F	F	T	T
(2)	T	F	F	T
(3)	F	T	T	F
(4)	T	T	F	F

150. Regarding recombinant DNA technology and gene expression, which of the following statements is **incorrect**?

- (1) DNA ligase joins the cut ends of DNA strands to form phosphodiester bonds.
- (2) Competent bacteria are treated with calcium chloride to take up plasmids.
- (3) Gel electrophoresis separates DNA fragments based on their molecular weight.
- (4) Bioreactors maintain optimal conditions only for bacterial culture.

151. Which of the following is a characteristic feature of adaptive/acquired immunity?

- (1) It provides immediate protection upon first exposure to a pathogen.
- (2) It generates memory cells that remember previous pathogenic encounters.
- (3) It is entirely non-specific in nature.
- (4) It works independently of the innate immune system.

152. Which of the following pairs is **correctly** matched?
- (1) *EcoRI* restriction enzyme → cuts GAATTC palindromic sequence producing sticky ends
 - (2) *Taq* polymerase → denatures DNA at high temperatures (95°)
 - (3) DNA ligase → removes nucleotides from DNA ends
 - (4) Reverse transcriptase → synthesizes double-stranded RNA from single-stranded DNA template

153. Match List-I with List-II

List-I		List-II	
(A)	Bioprocess engineering	(I)	Detecting HIV in early asymptomatic stage
(B)	PCR	(II)	Maintenance of sterile ambience in chemical engineering processes
(C)	ELISA	(III)	Identifying insertional inactivation
(D)	Blue-white screening	(IV)	Determining presence of antigens or antibodies

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
 - (2) A-I, B-II, C-III, D-IV
 - (3) A-III, B-II, C-IV, D-I
 - (4) A-II, B-I, C-IV, D-III
154. A 10-year-old child presents with recurrent infections and diagnosis confirms ADA deficiency. Which of the following is the most appropriate treatment?
- (1) Only enzyme replacement therapy for permanent cure
 - (2) Supportive care with antibiotics and awaiting natural immune recovery
 - (3) Bone marrow transplantation alone
 - (4) Gene therapy introducing functional ADA gene into stem cells at early embryonic stage

155. Which of the following statements **correctly** describe features of cancer cells?

- A. Cancer cells lose contact inhibition property.
- B. Cancer cells can undergo unlimited division.
- C. Malignant tumors have the ability of metastasis.
- D. Benign tumors can undergo metastasis like malignant tumors.
- E. Proto-oncogenes when normally express can cause cancer.

Select correct statements:

- (1) A, B, C and E only
- (2) A, B, C, D and E
- (3) A, C and E only
- (4) A, B and C only

156. Which of the following statements about gel electrophoresis is **incorrect**?

- (1) DNA moves toward the positive electrode because it is negatively charged
- (2) Smaller DNA fragments migrate faster through the gel matrix than larger fragments
- (3) Agarose gel provides a sieving effect based on pore size
- (4) Ethidium bromide intercalation stops DNA movement during electrophoresis

157. Given below are two statements.

Statement I: The origin of replication (ori) is not required for a cloning vector to replicate autonomously in host cells.

Statement II: Without ori sequence, the foreign DNA inserted into the vector cannot maintain itself through successive cell divisions.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

158. Match List-I with List-II

List-I		List-II	
(A)	Bone marrow	(I)	Site of T-cell maturation
(B)	Thymus	(II)	Filter of blood, stores RBCs
(C)	Spleen	(III)	Production of all blood cells
(D)	Lymph nodes	(IV)	Traps antigens from tissue fluid

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-III, B-I, C-II, D-IV
- (4) A-II, B-III, C-IV, D-I

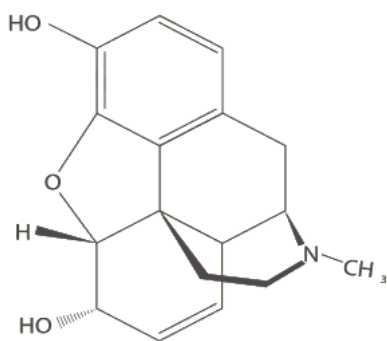
159. Read the following statements regarding golden rice and genetically modified crops:

- A. Golden rice has increased Vitamin A content due to transgene expressing carotenoid biosynthesis enzymes.
- B. GM crops have increased tolerance to abiotic stresses like drought and salt.
- C. *Bt* crops reduce reliance on chemical pesticides.
- D. All GM organisms produce identical beneficial traits regardless of transformation method.
- E. GM crops may require less fertilizer due to improved nutrient uptake efficiency.

Which of the above statements is/are **correct**?

- (1) A, B, C and E only
- (2) A, B, C, D and E
- (3) A, C and E only
- (4) B, C, D and E only

160. Choose **incorrect** statement regarding the drug of abuse represented by following chemical structure.



- (1) It interacts with specific opioid receptors present in the central nervous system and gastrointestinal tract.
 - (2) It is a very effective sedative and painkiller useful for patients who have undergone surgery.
 - (3) It is chemically diacetylmorphine, which is a white, odourless, and bitter crystalline compound.
 - (4) It is extracted from the latex of the poppy plant, *Papaver somniferum*.
161. A researcher has successfully cloned human growth hormone (hGH) gene in *E. coli* and wants to scale up production. The most important consideration for this scale-up is:
- (1) Using laboratory flasks to produce more product.
 - (2) Using stirred-tank bioreactors to maintain optimal temperature, pH, oxygen, and nutrient levels.
 - (3) Adding more antibiotics to prevent contamination.
 - (4) Increasing the number of restriction enzymes used during cloning.
162. Which of the following statements is **not** correct?
- (1) Allergies result from overreaction of immune system to harmless substances
 - (2) Allergic antibodies are of IgG type
 - (3) Histamine release from mast cells causes allergy symptoms
 - (4) Auto-immune diseases result from immune system attacking self-antigens

163. Match List-I with List-II.

List-I (PCR component)		List-II (Function)	
(A)	Taq polymerase	(I)	Provides DNA template
(B)	Primers	(II)	Catalyzes DNA synthesis
(C)	Deoxyribonucleotides	(III)	Bind to specific target sequences
(D)	Genomic DNA	(IV)	Building blocks for new DNA strands

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-I, B-IV, C-II, D-III
- (4) A-IV, B-I, C-III, D-II

164. A pregnant woman develops immunosuppression. Tests show low CD4⁺ T-cell count and high viral load. What is the most critical concern for her unborn child?

- (1) The child will definitely not develop AIDS at birth.
- (2) There is significant risk of HIV transmission through placenta.
- (3) The child's immune system will be stronger due to antibodies from mother.
- (4) HIV can only be transmitted through breast milk, not during pregnancy.

165. Which of the following statements is/are **true** about transformation and gene expression?

- A. Competent cells have increased permeability to take up exogenous DNA.
- B. Heat shock facilitates DNA uptake in chemical competence method.
- C. Selectable markers allow identification of transformed cells.

- (1) A and B only
- (2) A and C only
- (3) A, B and C
- (4) B and C only

166. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Transgenic mice constitute over 95% of all transgenic animals produced.

Reason R: Mice have long generation time, large size, well-characterized genome, and can be easily maintained in laboratory.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is true, but R is false.
- (4) A is false, but R is true.

167. Which of the following statements about the pBR322 plasmid vector is **not** correct?

- (1) It has multiple cloning sites for restriction enzyme cutting.
- (2) It cannot replicate independently in bacterial cells.
- (3) Insertional inactivation of tet^R gene indicates recombinant plasmid.
- (4) It has a specific ori site and a rop site for replication control.

- 168.** Which of the following about antibody structure is **correct**?
- (1) Antibodies consist of 2 heavy chains and 2 light chains.
 - (2) The antigen binding sites are located on the heavy chains only.
 - (3) Trisulphide bonds connect the heavy and light chains.
 - (4) All antibodies have identical variable regions.
- 169.** Which of the following statement about gene therapy is **correct**?
- (1) Gene therapy has successfully cured all genetic diseases in humans.
 - (2) Retroviral vectors integrate foreign DNA into host genome for expression.
 - (3) The first clinical gene therapy treated cystic fibrosis patients.
 - (4) Gene therapy cannot be applied to somatic cells.
- 170.** Given below are two statements:
Statement I: Primary immune response is slower than secondary immune response.
Statement II: Secondary response is faster because it has memory cells that recognize pathogens quickly.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
- 171.** In gel electrophoresis, which factor does **not** significantly affect DNA fragment separation?
- (1) Agarose concentration (pore size)
 - (2) Duration and voltage applied
 - (3) Size of the DNA fragments
 - (4) Type of restriction enzyme used for cutting
- 172.** Read the following statements about vaccination and immunisation programs:
- A. Vaccines contain antigens or weakened/inactivated pathogens.
 - B. Vaccination provides passive immunity.
 - C. Hepatitis B vaccine is produced using recombinant DNA technology.
 - D. Vaccination generates T lymphocytes only.
- Which of the above statements is/are **incorrect**?
- (1) A and B
 - (2) B and C
 - (3) B and D
 - (4) A and D
- 173.** Scientists want to develop disease-resistant rice varieties. Which of the following approach would be **most** suitable?
- (1) Classical hybridization with wild rice varieties.
 - (2) Introduction of resistance genes using *Agrobacterium*-mediated transformation.
 - (3) Chemical mutagenesis followed by selection.
 - (4) Waiting for natural mutations to arise.
- 174.** Which of the following statements is **not** true regarding extraction and purification of DNA?
- (1) Lysozyme is used to break down bacterial cell walls.
 - (2) Protease removes all nucleic acids from macromolecule mix.
 - (3) Ribonuclease removes RNA contamination.
 - (4) Purified DNA precipitates when ethanol is added.
- 175.** Which disease-prevention pairing is **incorrect**?
- (1) Malaria → Vector control and use of mosquito nets
 - (2) Typhoid → Food hygiene and water purification
 - (3) Pneumonia → Avoiding air-borne droplets and vaccination
 - (4) AIDS → Isolation and quarantine of infected persons only
- 176.** Read the following statements about downstream processing and product formulation:
- A. Downstream processing begins after the fermentation is complete.
 - B. Separation and purification involve methods like chromatography.
 - C. Formulation includes addition of preservatives and excipients.
 - D. Clinical trials are not required for therapeutic products.
 - E. Quality control testing ensures safety and efficacy.
- Select the option with **correct** statements only.
- (1) A, B, C and E only
 - (2) A, B, C, D and E
 - (3) A, C and E only
 - (4) B, C, D and E only
- 177.** Which of the following **correctly** describes the cell-mediated immune response?
- (1) It primarily involves production of antibodies by B cells.
 - (2) It is mediated by T-lymphocytes and does not involve antibodies.
 - (3) It provides immediate protection upon first exposure.
 - (4) It is ineffective against viral infections.

178. Which of the following is/are example of lifestyle related non infectious diseases?

- (1) Cancer
- (2) Obesity and diabetes
- (3) Auto-immune
- (4) Genetic disorders such as haemophilia

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

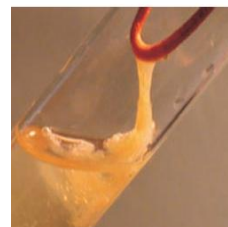
Assertion A: Somatic hybridization between tomato and potato protoplasts results in formation of pomato.

Reason R: Somatic hybridization allows for the combination of desired traits from two different species into a single plant through sexual reproduction.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is false, but R is true.
- (4) A is true, but R is false.

180. Which of the following phenomenon is represented by given diagram?



- (1) Elution
- (2) Culture
- (3) Spooling
- (4) Filtration



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>

Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____